

WILLIAMS LAKE, BC, Canada

WMO: 711040

Lat: 52.1862N

Lon: 122.0656W

Elev: 940

StdP: 90.53

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-26.5	-22.7	-30.9	0.2	-26.2	-26.9	0.4	-22.5	13.1	0.8	11.6	0.8	1.4	140	0.627

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	28.3	15.1	26.3	14.4	24.4	13.6	15.9	25.6	15.1	24.3	14.3	22.8	2.9	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.7	10.2	17.0	11.5	9.5	16.5	10.6	8.9	15.9	47.8	25.6	45.2	24.4	42.9	22.9	19.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.9	8.4	7.4	DB	-30.9	31.3	4.7	1.7	-34.2	32.5	-37.0	33.5	-39.6	34.5	-42.9	35.8	(4)
(5)			WB	-31.1	17.5	4.6	1.0	-34.4	18.3	-37.1	18.9	-39.7	19.5	-43.0	20.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.7	-6.3	-4.7	0.0	4.7	9.8	13.2	16.2	15.5	10.8	4.6	-1.8	-6.5	(6)	
	DBStd	9.55	7.74	6.57	5.53	3.69	3.92	3.24	3.34	3.34	3.88	4.07	5.90	6.82	(7)	
	HDD10.0	2533	506	412	311	164	53	6	1	2	36	174	356	513	(8)	
	HDD18.3	5023	764	646	568	409	266	157	84	98	226	427	606	771	(9)	
	CDD10.0	584	0	0	0	5	46	102	192	173	61	6	0	0	(10)	
	CDD18.3	34	0	0	0	0	1	3	18	12	1	0	0	0	(11)	
	CDH23.3	642	0	0	0	0	22	76	287	219	38	1	0	0	(12)	
	CDH26.7	142	0	0	0	0	2	16	73	47	5	0	0	0	(13)	
(14) Wind	WSAvg	2.7	3.0	2.9	3.2	3.0	2.5	2.3	2.1	2.0	2.3	3.0	3.3	3.3	(14)	
(15) Precipitation	PrecAvg	451	40	26	21	24	38	49	54	46	38	35	37	44	(15)	
	PrecMax	670	123	56	67	66	79	125	201	127	97	101	94	87	(16)	
	PrecMin	258	7	7	5	2	8	14	6	8	5	6	6	8	(17)	
	PrecStd	98	27	13	13	16	20	28	36	31	24	24	19	20	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.8	8.4	14.4	20.8	25.7	28.7	31.3	30.5	27.0	19.0	11.5	6.1	(19)	
		MCWB	3.4	3.4	6.4	10.0	12.5	15.0	16.2	15.5	13.8	10.8	6.7	2.9	(20)	
	2%	DB	4.8	6.0	11.1	16.7	23.0	25.6	28.7	27.8	24.0	15.5	9.0	4.1	(21)	
		MCWB	2.3	2.0	4.7	8.2	11.5	13.6	15.3	14.9	13.2	9.3	5.7	1.3	(22)	
	5%	DB	3.5	4.5	9.0	14.1	20.5	22.9	26.6	25.9	21.4	13.0	6.8	2.7	(23)	
		MCWB	1.4	1.2	3.6	6.7	10.6	12.5	14.6	14.6	12.5	8.1	3.7	0.5	(24)	
	10%	DB	2.2	3.0	7.1	11.7	17.9	20.5	24.4	23.7	18.6	11.0	4.8	1.3	(25)	
		MCWB	0.3	0.4	2.4	5.3	9.7	11.8	13.9	13.8	11.3	6.7	2.7	-0.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.0	4.1	7.0	10.4	13.5	15.8	17.1	17.3	15.0	11.7	7.6	3.5	(27)	
		MCDB	6.1	7.3	13.7	19.3	23.4	25.5	28.4	26.2	24.6	17.4	10.5	5.6	(28)	
	2%	WB	2.6	2.6	5.2	8.8	12.2	14.5	16.0	16.0	13.9	10.0	5.9	1.7	(29)	
		MCDB	4.4	5.3	10.3	15.8	20.8	23.4	25.9	25.6	22.2	14.5	8.7	3.5	(30)	
	5%	WB	1.5	1.5	4.0	7.2	11.2	13.3	15.2	15.0	12.8	8.5	4.2	0.7	(31)	
		MCDB	3.2	4.0	8.1	13.2	19.0	21.0	24.6	24.0	20.2	12.6	6.2	2.5	(32)	
	10%	WB	0.5	0.5	2.8	5.8	10.2	12.3	14.4	14.2	11.7	7.0	2.8	-0.3	(33)	
		MCDB	1.9	2.8	6.6	11.1	16.7	19.3	23.0	22.4	18.0	10.3	4.7	1.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.9	8.8	9.6	11.0	11.9	11.3	12.2	13.0	12.0	9.2	6.5	6.3	(35)	
		MCDBR	6.3	8.8	11.2	14.8	16.3	15.7	16.4	17.1	16.9	12.5	7.4	5.7	(36)	
	5% WB	MCWBR	4.6	5.8	6.6	8.2	7.6	6.7	6.3	7.1	8.2	7.4	5.1	4.4	(37)	
		MCWBR	6.0	8.2	10.6	13.5	14.3	13.5	14.5	15.3	15.3	11.6	6.9	5.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.258	0.266	0.288	0.313	0.333	0.334	0.325	0.327	0.300	0.286	0.270	0.259	(40)	
		taud	2.414	2.442	2.491	2.467	2.453	2.475	2.539	2.523	2.603	2.612	2.528	2.362	(41)	
	Ebn at Noon		774	874	919	928	920	919	922	903	897	846	760	696	(42)	
		Edh at Noon	55	72	84	98	105	104	96	92	76	61	50	49	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.90	1.87	3.20	4.54	5.32	5.68	5.98	5.09	3.50	2.02	0.95	0.67	(44)	
	RadStd		0.08	0.17	0.22	0.33	0.37	0.47	0.56	0.24	0.33	0.18	0.10	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page